Supplementary Material for

Modal interactions in strongly nonlinear coupled oscillators with closely spaced modes and non-classical damping distribution

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This supplementary document provides the derivation of the action-angle fromulation of the nonlinear degree of freedom (DoF) used in Section 4 of the main paper.

Action–angle variables of the nonlinear DoF

In the uncoupled conservative case, the nonlinear DoF is an undamped and unforced Duffing oscillator governed by,

$$\ddot{x}_1(t) + \alpha x_1^3(t) = 0 \tag{1}$$

with corresponding action computed as,

$$I_1 = \frac{2}{\pi}\int_0^{x_1^\star} \sqrt{2\left(E_0 - \frac{\alpha}{4}x_1^4\right)}\mathrm{d}x_1 = \frac{2\sqrt{\alpha}}{3\pi}x_1^{\star 3}K\left(m\right) \tag{2}$$

where $x_1^\star = \left(\frac{4}{\alpha}E_0\right)^{\frac{1}{4}}$, E_0 is the constant energy level of (1), and K is the complete elliptic integral of the first kind with an elliptic parameter $m = 1/2$. It follows that,

$$x_1^\star = \left(\frac{1}{4\alpha}\right)^{1/6}\left(\frac{3\pi}{K\left(m\right)}\right)^{1/3} I_1^{1/3} \equiv \Lambda I_1^{1/3} \tag{3}$$

which gives the energy-action relationship:

$$E_0 = H_1(\dot{x}_1(t), x_1(t)) = \frac{\alpha}{4}\Lambda^4 I_1^{4/3} = \text{const.} \tag{4}$$

Thus, unlike in the linear system, where energy and action are linearly proportional, the energy of the Duffing oscillator (1) scales nonlinearly with the action.

The response of the Duffing oscillator (1) can be expressed in terms of the Jacobi elliptic cosine function as,

$$x_1(t) = x_1^\star \mathrm{cn}(\Omega t, m) \tag{5}$$

from which its frequency Ω is:

$$\Omega = \sqrt{\alpha}x_1^\star = \left(\frac{3\pi\alpha}{2K\left(m\right)}\right)^{1/3} I_1^{1/3}(t) \tag{6}$$

Since the Jacobi elliptic function $\mathrm{cn}(\Omega t, m)$ has a period equal to $4K(m)$, the physical period of oscillation is given by,

$$T = \frac{4K(m)}{\Omega} \tag{7}$$

from which the angle variable $\theta_1(t)$ of Eq. (1) is expressed as:

$$\theta_1(t) = \frac{2\pi}{T}t = \left(\frac{3\pi^4\alpha}{16K^4\left(m\right)}\right)^{1/3} I_1^{1/3}t = \omega_1(I_1)t \tag{8}$$